

Hamzah Zuhrah

Cybersecurity Student at ODU | IT Help Desk Experience | Seeking SOC & Security Operations Internships
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PROFESSIONAL SUMMARY

Cybersecurity student at Old Dominion University (B.S. expected May 2027, GPA 3.95) and Virginia Tech Industrial Engineering graduate with hands-on IT help desk, secure systems, and compliance-oriented operations experience. Experienced supporting user access, regulated workflows, incident documentation, security controls, network fundamentals, Python, Linux, cloud security, and technical documentation. Seeking SOC, security operations, cybersecurity analyst, GRC, compliance, vulnerability management, cloud security, or junior security engineering opportunities.

TECHNICAL SKILLS

Security: Incident response, network security, threat modeling, vulnerability assessment, access control, system hardening, phishing analysis, cryptography, NIST CSF 2.0, NIST SP 800-61r3, security documentation

GRC & Compliance: Audit readiness, policy and procedure adherence, evidence handling, confidentiality, role-based access, regulated system use, risk identification, incident reporting, clearance-sensitive workflows

Systems & Networking: Linux/Kali Linux, Windows, macOS, TCP/IP, DNS, DHCP, routing and switching fundamentals, Nmap, packet analysis, Wireshark, virtual machines

Programming & Tools: Python, Bash, Git, GitHub, automated testing, secure protocol design, structured audit logging, TLS 1.2+, AES-256-GCM, scrypt

State / Justice Systems: OPUS, OMNIXX, DCI, and multiple clearance-controlled state systems

Cloud: AWS IAM, EC2, S3, cloud architecture, encryption, monitoring, and cloud security fundamentals

SELECTED CYBERSECURITY PROJECT

Secure File Transfer System v2

2026

- Re-engineered a Python socket prototype into a security-focused file exchange platform using client-side AES-256-GCM encryption, verified TLS 1.2+, salted scrypt authentication, and per-user storage boundaries.
- Designed a versioned length-prefixed protocol with rate limiting, structured audit logs, path safety controls, atomic transfer verification, and cleanup of partial files.
- Validated confidentiality, integrity, authentication, traversal rejection, tamper detection, wrong-passphrase rejection, failed-transfer cleanup, and TLS behavior with automated security-focused tests.

APPLIED SECURITY RESEARCH

Independent Research Opportunity - COVA CCI

Ongoing

- Researching encrypted DNS security and privacy, with a focus on how DoH/DoT adoption affects user privacy, network visibility, performance, and defenders' ability to detect DNS-based threats.
- Planning a Wireshark-supported demonstration comparing traditional DNS visibility with encrypted DNS behavior and documenting implications for security monitoring, acceptable-use controls, and enterprise policy.

PROFESSIONAL EXPERIENCE

Information Technology Trainee | Gateway Homes, Inc.

Jan 2026 - Present

Remote | Apprenticeship

- Provide Tier 1 technical support for staff, including workstation setup, printer troubleshooting, software updates, account access, password resets, and routine system checks.
- Document technical issues, escalate higher-risk or complex incidents to the IT team, and support reliable day-to-day operations for remote and office users.
- Maintain hardware/software inventory and assist with user education, basic security hygiene, and consistent support procedures.

NCDAC | Public Safety & Clearance-Controlled Systems Operations

- Use OPUS, OMNIXX, DCI, and multiple clearance-controlled state systems while following confidentiality, access-control, identity verification, and records-handling requirements.
- Support compliance-driven operations by protecting sensitive information, maintaining accurate system records, following state policy, and escalating irregularities through appropriate channels.
- Apply security-minded judgment in regulated workflows involving restricted systems, role-based access, auditability, incident awareness, and procedural compliance.

Intern - Industrial Systems Engineering | Moog Space and Defense

May 2025 - Dec 2025

Internship

- Supported engineering tasks, technical coordination, and cross-functional project teams in a defense-oriented industrial systems environment.
- Contributed to continuous-improvement and business-process initiatives with attention to process reliability, documentation, and operational risk.
- Assisted project-engineering activities and collaborated with multidisciplinary teams to support reliable, efficient operations.

Residential Supervisor | NeuroRestorative

Jan 2022 - May 2025

Blacksburg, Virginia

- Promoted through progressively responsible roles from Life Skills Trainer to Medication Administrator, Team Leader, and Residential Supervisor.
- Supported compliance and audit readiness while coordinating teams, hiring, schedules, budgets, medication documentation, incident reports, and operational logistics.
- Handled sensitive resident information and regulated procedures with confidentiality, accuracy, and escalation discipline.

APPLIED SECURITY TRAINING

Cybersecurity Lab Practice (Self-Directed)

Sep 2025 - Present

- Conduct hands-on labs in a virtualized Kali Linux environment, including Nmap scanning, enumeration, vulnerability identification, file permissions, user management, security configuration, and system reconnaissance.

Cloud Computing Student Project

Feb 2026 - Present

- Complete guided AWS labs covering IAM, EC2, cloud storage, access control, encryption, monitoring, architecture, deployment models, and cloud security fundamentals.

EDUCATION

Old Dominion University | B.S. Cybersecurity | Jan 2026 - May 2027 (Expected) | GPA: 3.95

Virginia Tech | Bachelor of Engineering, Industrial Engineering | Aug 2021 - May 2025

Northern Virginia Community College | Associate of Applied Science

CREDENTIALS

Professional development: Preparing for CompTIA Security+ (SY0-701)

Additional credentials: Registered Behavior Technician, BACB (expires Jan 2027); Qualified Brain Injury Support Provider, NeuroRestorative (expires May 2027)